

# DeepSeek-R1

## DeepSeek-R1

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### Demonstration Environment

**Development Board:** Jetson Orin series motherboard

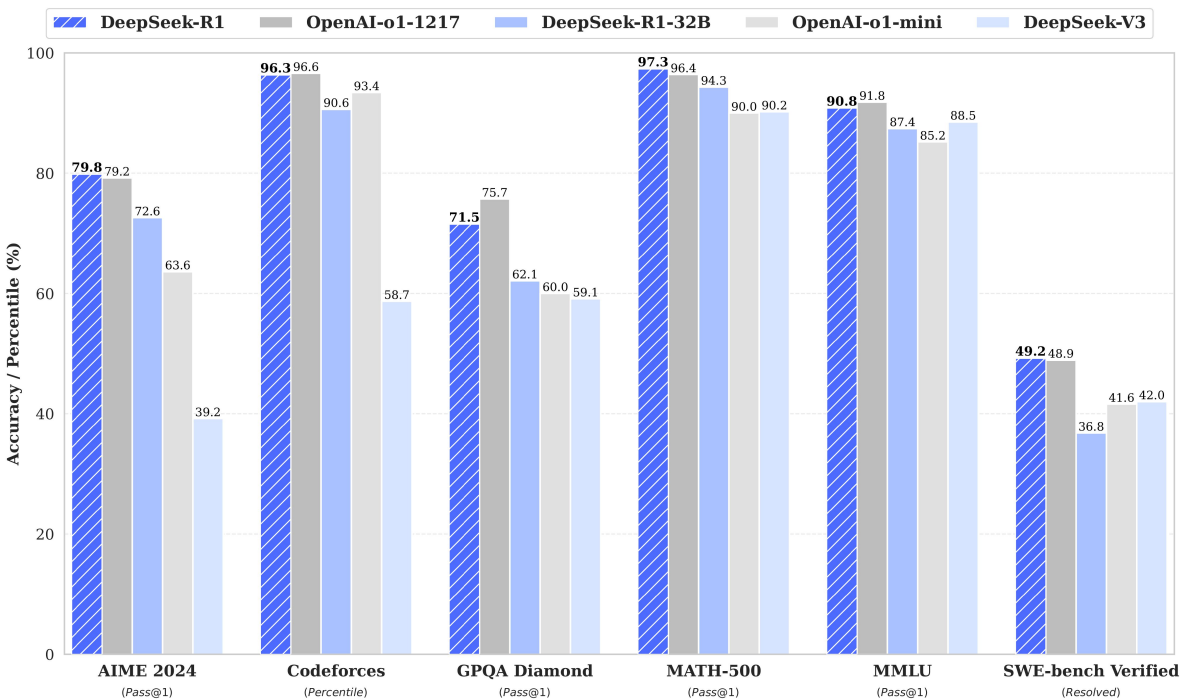
**Note:** Due to performance limitations, the Jetson Orin Nano 4GB requires the reduced-parameter version

DeepSeek-R1 is an open-source Large Language Model (LLM) designed by DeepSeek for understanding and generating code.

## 1. Model Size

| Model            | Size  |
|------------------|-------|
| deepseek-r1:1.5b | 1.1GB |
| deepseek-r1:7b   | 4.7GB |
| deepseek-r1:8b   | 5.2GB |

## 2. Performance



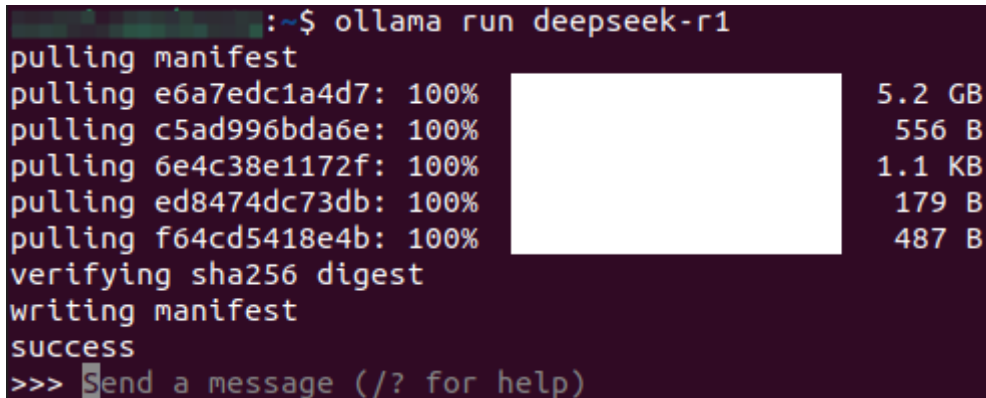
## 3. Using DeepSeek-R1

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### 3.1 Running DeepSeek-R1

Use the run command to start running the R model. If the model is not downloaded, the model from the Ollama model library will be automatically downloaded:

```
ollama run deepseek-r1
```

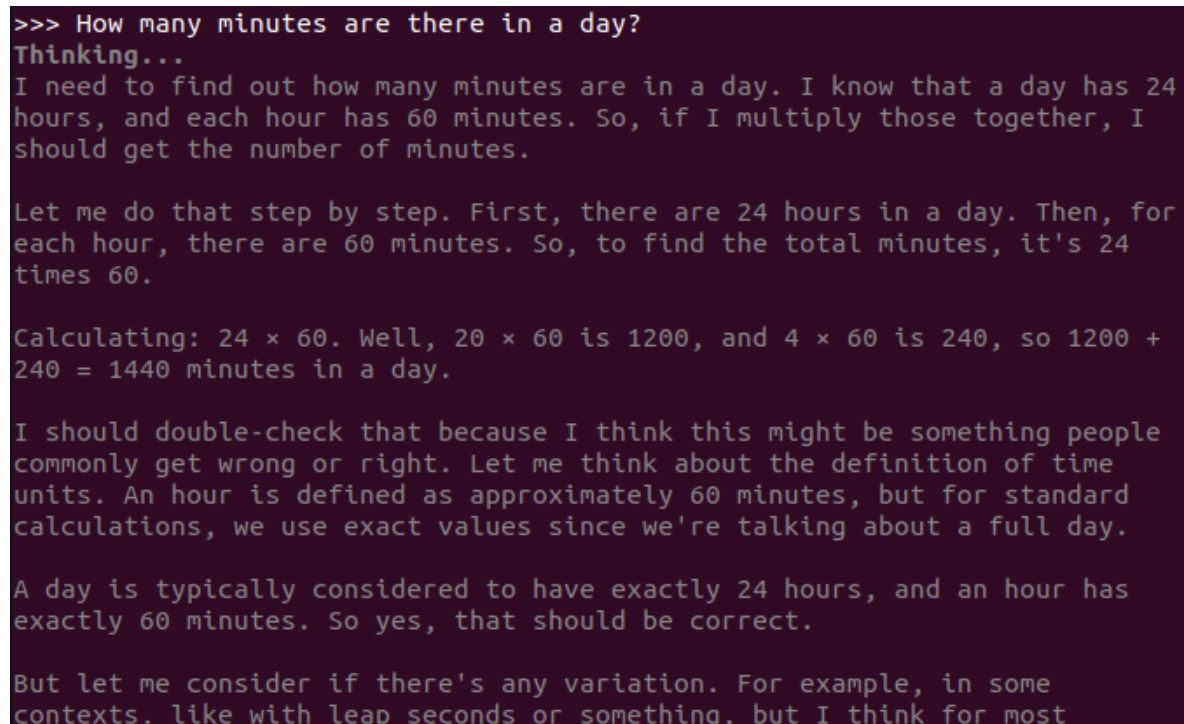


```
~$ ollama run deepseek-r1
pulling manifest
pulling e6a7edc1a4d7: 100% 5.2 GB
pulling c5ad996bda6e: 100% 556 B
pulling 6e4c38e1172f: 100% 1.1 KB
pulling ed8474dc73db: 100% 179 B
pulling f64cd5418e4b: 100% 487 B
verifying sha256 digest
writing manifest
success
>>> Send a message (/? for help)
```

### 3.2 Starting a Conversation

```
How many minutes are there in a day?
```

Response time depends on your hardware configuration, so please be patient!



```
>>> How many minutes are there in a day?
Thinking...
I need to find out how many minutes are in a day. I know that a day has 24
hours, and each hour has 60 minutes. So, if I multiply those together, I
should get the number of minutes.

Let me do that step by step. First, there are 24 hours in a day. Then, for
each hour, there are 60 minutes. So, to find the total minutes, it's 24
times 60.

Calculating: 24 x 60. Well, 20 x 60 is 1200, and 4 x 60 is 240, so 1200 +
240 = 1440 minutes in a day.

I should double-check that because I think this might be something people
commonly get wrong or right. Let me think about the definition of time
units. An hour is defined as approximately 60 minutes, but for standard
calculations, we use exact values since we're talking about a full day.

A day is typically considered to have exactly 24 hours, and an hour has
exactly 60 minutes. So yes, that should be correct.

But let me consider if there's any variation. For example, in some
contexts, like with leap seconds or something, but I think for most
```

### 3.3 Ending the Conversation

Use the `ctrl+d` shortcut or `/bye` to end the conversation!

## References

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Ollama

Official Website: <https://ollama.com/>

GitHub: <https://github.com/ollama/ollama>

DeepSeek-R1

Ollama Model: <https://ollama.com/library/deepseek-r1>

GitHub: <https://github.com/deepseek-ai/DeepSeek-r1>